

## **Computer Network Lab Questions (BIT)**

### **1. IP Address Configuration**

- Configure a static IP address on a Windows
- Verify the configuration using ipconfig (Windows)

### **2. Network Connectivity Testing**

- Use the ping command to test connectivity between two computers.
- Record and analyze the results.

### **3. Traceroute Experiment**

- Use tracert (Windows) to trace the path to a website.
- Identify the number of hops.

### **4. Peer to Peer connection**

### **5. Basic Network Setup in cisco packet tracer**

Objective: Connect two PCs using a switch.

Tasks:

1. Place 2 PCs and 1 Switch.
2. Connect devices using copper straight-through cables.
3. Assign IP addresses:
  - PC0: 192.168.1.10/24
  - PC1: 192.168.1.20/24
4. Test connectivity using ping.

### **6. Network Cable Creation**

- Prepare a straight-through Ethernet cable using RJ-45 connectors.
- Test the cable using a cable tester.

### **7. Switch Configuration in cisco packet tracer**

- Configure a switch with:

- Hostname
- Management IP address
- Password protection

## **8. VLAN Configuration in cisco packet tracer**

- Create two VLANs on a switch.
- Assign ports to different VLANs.
- Test communication between VLANs.

## **9. Router Configuration**

- Configure basic settings on a router:
  - Hostname
  - Password
  - Interface IP address
  - Save configuration

## **10. Inter Vlan connection using router in cisco packet tracer**

Objective: Enable communication between VLANs.

Tasks:

1. Create VLANs.
2. Configure trunk port.
3. Configure Router-on-a-Stick.
4. Test communication.

## **11. Configure Router in Stick Model in cisco packet tracer**

- Pass and test connection for 4 vlan through single port of Router

## **12. DHCP service Configuration in cisco packet tracer**

- Configure a DHCP server.
- Verify automatic IP assignment to clients.

### **13. DNS Testing**

- Use nslookup to find the IP address of a domain.
- Explain DNS resolution.

### **14. Configure network device using console cable**

### **15. Configure SSH Connection**

Objective: Secure remote access to a router.

Tasks:

1. Configure hostname and domain name.
2. Generate RSA keys.
3. Create local user.
4. Enable SSH.

### **16 Static Routing Configuration in cisco packet**

Objective: Connect three networks using two routers.

Tasks:

1. Create three LANs.
2. Configure router interfaces.
3. Configure static routes.
4. Verify end-to-end connectivity.

### **17 . RIP Routing Configuration in cisco packet tracer**

Objective: Configure RIP v2.

Tasks:

1. Connect 3 routers.
2. Assign IP addresses.

3. Configure RIP v2.

## **18. OSPF Routing Configuration in cisco packet tracer**

### **19. NAT Configuration**

- Configure Network Address Translation on a router.
- Verify internet access from private IP addresses.

### **20. Firewall Configuration**

- Configure firewall rules to block and allow specific services.
- Test the results.

### **21. Network Troubleshooting**

Using commands such as:

- ping
- ipconfig
- netstat
- nslookup
- tracert

Diagnose and solve a given network connectivity problem.

### **22. Design the topology of Namuna College Lab and configure.**

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